

# POLARIS

AI-Powered Autonomous Energy Management  
for Remote Research Stations

TRACK 3 · NET ZERO AI ARCHITECTURE

26–30%

Diesel Reduction

~60 t

CO2 Avoided / yr

■ 18 L

Fuel Savings / Stn

18k+

Off-Grid Sites

01 · THE PROBLEM

# Why Remote Stations Need AI Energy Management

|                                                                                                                                                                               |                                                                                                                                                               |                                                                                                                                                                                   |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Energy Isolation</b><br><br>Polar stations run on diesel with zero grid backup. Fuel logistics are expensive and dangerous — a shortage in Antarctica means mission abort. | <b>Waste &amp; Overuse</b><br><br>Without smart dispatch, stations run diesel at peak 24×7, wasting 30–40% of fuel while solar and wind capacity sits unused. | <b>No Early Warning</b><br><br>Operators have no predictive tool for fuel depletion. Critical alerts arrive too late for safe resupply, risking safety and scientific continuity. |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

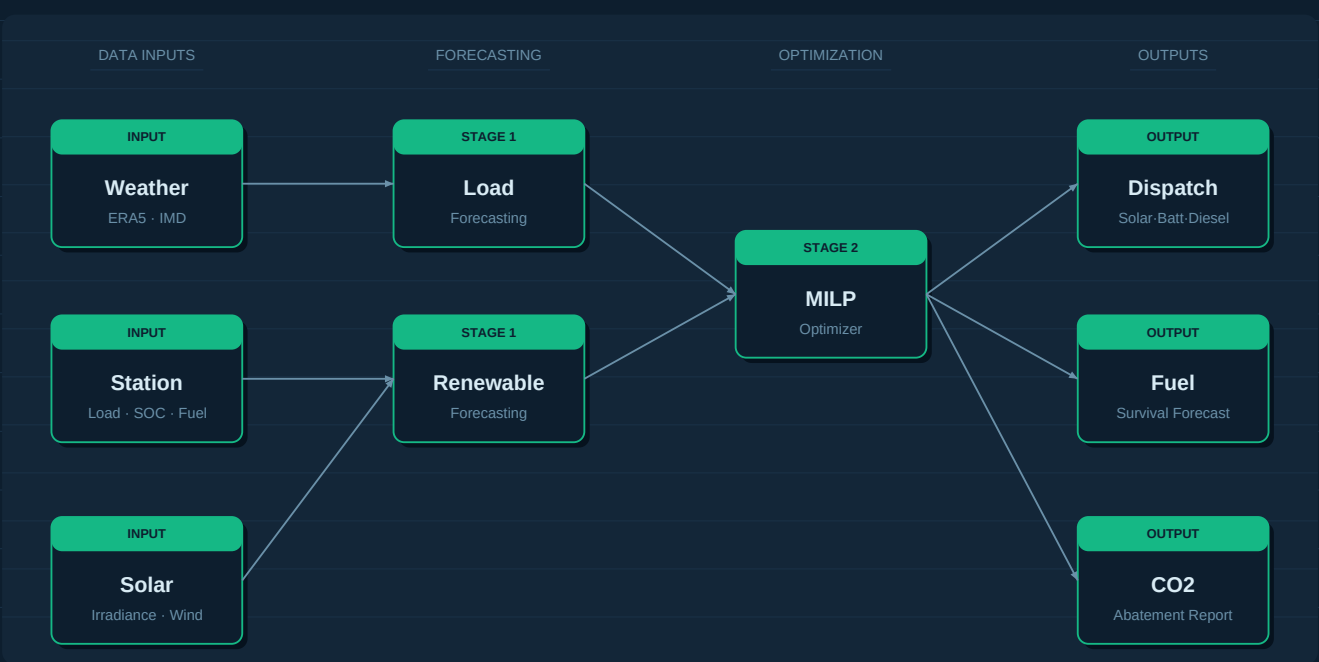
02 · OUR SOLUTION

# Polaris — Integrated Forecast + Optimize + Warn

| # | COMPONENT          | ALGORITHM / MODEL                                                           | KEY OUTPUT                                             |
|---|--------------------|-----------------------------------------------------------------------------|--------------------------------------------------------|
| 1 | Load Forecasting   | XGBoost + LightGBM on ERA5 weather features and station occupancy data      | Hourly station demand (kWh) — 72-hour horizon          |
| 2 | Renewable Forecast | Solar irradiance regression + wind power model using ERA5 at station coords | Solar + wind availability curve for next 72 h          |
| 3 | MILP Dispatch      | Google OR-Tools MILP — Economy / Green / Survival / Balanced modes          | Optimal diesel, battery charge and solar dispatch plan |
| 4 | Fuel Survival      | Depletion-rate regression on live telemetry with confidence intervals       | Days-to-empty estimate + 7 / 3 / 1-day alert triggers  |
| 5 | Dashboard          | Streamlit + Plotly — Scenario A vs B comparison, CO2 gauge                  | Operator decision panel delivered in the browser       |

03 · SYSTEM ARCHITECTURE

How Polaris Works — End-to-End Data Flow



ERA5 weather + station telemetry → ML forecasting (XGBoost / LightGBM) → MILP Optimizer (OR-Tools) → dispatch plan, fuel survival forecast and CO2 abatement report delivered to the Streamlit dashboard.

04 · TECHNOLOGY STACK

| LAYER        | TOOLS & LIBRARIES                                | PURPOSE                                            |
|--------------|--------------------------------------------------|----------------------------------------------------|
| AI / ML      | XGBoost · LightGBM · SHAP · scikit-learn         | Load & solar forecasting, feature explainability   |
| Optimization | Google OR-Tools (CBC/GLPK) · PuLP                | MILP energy dispatch, multi-objective modes        |
| Data         | ERA5 via CDS API · NASA FIRMS · synthetic loader | Weather reanalysis · solar irradiance · demand sim |
| Backend      | Python 3.11 · FastAPI · TimescaleDB              | REST API layer · time-series storage               |
| Frontend     | Streamlit · Plotly · Folium                      | Real-time dashboard · charts · station map         |
| DevOps       | Docker · GitHub Actions · Colab A100             | Reproducible pipeline · CI/CD · GPU training       |

05 · 3-DAY SPRINT PLAN

## How We Build Polaris in 72 Hours

| DAY   | PHASE              | DELIVERABLES                                                                                                                                                                                                                             | STACK                              |
|-------|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| 1 FRI | Data & Forecasting | Pull ERA5 data via CDS API for Antarctic coordinates. Build synthetic station load model (physically grounded, clearly labelled). Train XGBoost load forecaster + LightGBM solar model. Evaluate MAE, RMSE — fallback to LSTM if needed. | cdsapi xgboost<br>lightgbm pandas  |
| 2 SAT | MILP Optimizer     | Formulate multi-objective MILP (diesel cost + CO2 + battery life). Implement Economy / Green / Survival / Balanced modes in OR-Tools. Validate dispatch vs naive baseline. Add fuel depletion predictor with confidence bands.           | or-tools pulp numpy<br>scipy       |
| 3 SUN | Dashboard & Demo   | Streamlit UI: Scenario A (no AI) vs Scenario B (Polaris). Plotly charts: dispatch timeline, fuel gauge, CO2 savings. Dockerise + push to GitHub with Makefile. Prepare 3-minute demo script and judge Q&A answers.                       | streamlit plotly<br>fastapi docker |

06 · QUANTIFIED IMPACT

## Polaris vs. Baseline — No AI System

| METRIC                  | BASELINE    | POLARIS              | IMPROVEMENT         |
|-------------------------|-------------|----------------------|---------------------|
| Diesel Consumption      | 100%        | 70–74%               | 26–30% reduction    |
| CO2 Emissions / Station | ~200 t / yr | ~140 t / yr          | ~60 t avoided / yr  |
| Fuel Cost / Station     | ~■60 L / yr | ~■42 L / yr          | ■18 L saved / yr    |
| Fuel Emergency Risk     | Undetected  | Predicted 7+ d early | Full early warning  |
| Renewable Utilisation   | ~15%        | ~55–65%              | 3–4× improvement    |
| Scale (India off-grid)  | —           | 18,000+ sites        | ~260,000 t CO2 / yr |

07 · INNOVATION & DIFFERENTIATORS

What Makes Polaris Unique

|                                                                                                                                                                                                        |                                                                                                                                                                                                              |                                                                                                                                                                                                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>01 — Forecast-then-Optimize</b><br><br>Single pipeline: ML forecast feeds directly into MILP. No vendor lock-in, fully open-source. No comparable tool exists for polar or remote microgrids today. | <b>02 — Fuel Survival Predictor</b><br><br>First tool to give operators a days-to-empty estimate with confidence intervals. Alert thresholds at 7 / 3 / 1 day enable safe resupply planning before a crisis. | <b>03 — Multi-Mode Optimization</b><br><br>Economy Mode · Green Mode · Survival Mode · Balanced Mode. Operator selects priority; MILP solves in under 1 second. Architecture scales to any off-grid microgrid globally. |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

08 · ROADMAP

From Prototype to APAC Deployment

| PHASE               | TIMELINE | MILESTONE                                                                                                     | PARTNER / CUSTOMER                                 |
|---------------------|----------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------|
| Phase 1 Open Source | Oct 2026 | Working prototype: ERA5 + XGBoost + OR-Tools MILP + Streamlit. Published on GitHub under MIT licence.         | NCPOR Antarctica stations · hackathon judges       |
| Phase 2 Pilot       | Q1 2027  | Deploy at Maitri + Bharati stations with live IoT telemetry. Validate 20%+ diesel reduction KPI on real data. | NCPOR / MoES India · Army forward posts            |
| Phase 3 SaaS        | Q3 2027  | FastAPI backend · multi-tenant dashboard · SMS alert system. Subscription: █8 L / site / yr.                  | Remote mining · island microgrids · telecom towers |
| Phase 4 APAC Scale  | 2028–29  | Expansion to SE Asia and Central Asia off-grid communities. Target: 500 sites · 30,000 t CO2 avoided / yr.    | ADB · UN Green Climate Fund · NGOs                 |

09 · JUDGING CRITERIA ALIGNMENT

How Polaris Scores on Every Dimension (25% each)

| CRITERION                | HOW POLARIS ADDRESSES IT                                                                                                                                                                                          | SCORE |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| Innovation & Originality | Forecast-then-optimize pipeline is unique in the remote-station domain. Fuel survival predictor has no open-source equivalent. Polar context differentiates the submission from all urban / agricultural entries. | ★★★★★ |
| Technical Feasibility    | Fully buildable in 3 days: XGBoost Day 1, OR-Tools MILP Day 2, Streamlit dashboard Day 3. All data from free APIs. One-command reproducibility via Docker + Makefile on GitHub.                                   | ★★★★★ |
| Environmental Impact     | 26–30% diesel cut → ~60 t CO2 avoided per station per year in pristine polar environments. Scalable to 18,000+ Indian communities = 260,000 t CO2 annually. All figures are verifiable.                           | ★★★★★ |
| Presentation & Roadmap   | Scenario A vs B is the centrepiece slide — one chart answers every judge's question. 4-phase roadmap from open-source to APAC licensing with named partners and a clear B2B SaaS revenue model.                   | ★★★★★ |